Sales Performance & Profitability Analysis

POWER BI PORTFOLIO PROJECT | RETAIL BUSINESS INTELLIGENCE

End-to-end Business Intelligence | Power BI Desktop | DAX | Data Modelling | Interactive Dashboards

TOTAL SALES	TOTAL PROFIT	PROFIT MARGIN
£989.58M	£425.33M	42.98%

Figures are approximate values from the Executive Overview screenshot. Verify against the final PBIX before publishing.

1. Project Overview

Project type: End-to-End Business Intelligence and Retail Sales Analytics

Tools: Microsoft Power BI Desktop, Power Query, DAX, Data Modelling, Data Visualisation and Performance Analyzer

Dataset: Approximately 1 million retail sales transactions, supported by product, customer, store and date information.

Developed an interactive, multi-page Power BI report to analyse retail sales performance, profitability, product performance, regional trends and customer behaviour. The project transforms transactional retail data into meaningful business insights using a star-schema data model, DAX calculations and interactive dashboards. Advanced functionality includes drill-through analysis, custom tooltips, bookmarks, dynamic field parameters and a What-If Discount Simulator.

2. Business Problem and Objectives

A large retail business needs a consolidated view of performance instead of relying on separate spreadsheets and manual calculations. The report addresses these questions:

- How much revenue and profit are we generating?
- Which products and categories contribute most to sales?
- Which regions and sales channels perform best?
- Which customer segments contribute the most revenue?
- How do sales and profitability change over time?
- What happens to profit if an additional discount is introduced?

3. Dataset and Data Model

Table	Purpose
Sales	Transaction-level sales, costs, profit, discounts and quantity
Products	Product names, categories and subcategories

Customers	Customer segments, regions and joining dates
Stores	Store names, regions and sales channels
Date	Year, quarter, month and date analysis
Extra Discount	Disconnected parameter table for What-If Analysis

The Sales fact table is linked to Customers, Products, Stores and Date dimension tables using one-to-many relationships. The disconnected Extra Discount parameter supports scenario calculations without directly filtering the Sales table. The star schema simplifies filtering and supports consistent DAX calculations.

4. Data Preparation and Transformation

The dataset was organised for analysis by checking data types, validating key columns and arranging the fact and dimension tables. The model supports reporting by product, customer, store, date and sales channel. Describe only data-cleaning steps actually performed when presenting the project.

5. Key DAX Calculations

Measure	Business meaning
Total Sales	Total revenue generated
Total Profit	Revenue remaining after recorded costs
Total Cost	Cost associated with sales
Profit Margin %	Profit as a percentage of sales
Total Orders	Distinct transaction count
Total Quantity	Total units sold
Sales YTD	Year-to-date revenue
Sales Previous Year	Prior-year comparable sales
Sales YoY Growth %	Year-over-year percentage change

```
Profit Margin % =
DIVIDE([Total Profit], [Total Sales], 0)
```

Profit margin complements revenue by showing how much of each pound of sales remains as profit. Time-intelligence measures support comparisons over time.

6. Dashboard Pages and Their Purpose

Page 1 — Executive Overview

Overall view of sales, profit, margin, monthly trends, leading products and sales channels.

Page 2 — Product Analysis

Compares products and categories by revenue, profit and profitability.

Page 3 — Regional Performance

Evaluates performance across geographic regions and sales channels.

Page 4 — Customer Insights

Analyses customer segments, purchasing activity and revenue contribution.

Page 5 — Dynamic Performance

Uses Field Parameters to switch between metrics in the same visual.

Page 6 — Discount Simulator

Explores additional discounts from 0% to 20% and their effect on simulated sales, profit and margin.

7. Advanced Power BI Features

Feature	Implementation	Business value
Drill-through	Product Deep Dive	Detailed product investigation
Custom tooltips	Category tooltip page	Additional context on hover
Bookmarks	Sales / Profit switching	Alternate views with buttons
Field Parameters	Dynamic Performance page	Metric switching without duplicate charts
What-If parameter	Extra Discount slider	Scenario analysis
DAX measures	Revenue, costs, margins	Filter-responsive calculations
Performance Analyzer	Visual timings	Identify bottlenecks

Discount Simulator — DAX

```
Simulated Sales =
VAR Discount = MAX('Extra Discount'[Extra Discount])
RETURN [Total Sales] * (1 - Discount)

Simulated Profit = [Simulated Sales] - [Total Cost]

Simulated Profit Margin % =
DIVIDE([Simulated Profit], [Simulated Sales], 0)

Profit Impact = [Simulated Profit] - [Total Profit]
```

The simulator uses the upper selected value from a range slider. It assumes unchanged sales volume and costs and is therefore a scenario analysis, not a demand forecast.

8. Performance Optimisation and Validation

Visual	Approximate recorded duration
Monthly Sales Trend	134 ms
Revenue by Sales Channel	404 ms
Sales vs Total Profit by Month	402 ms
Top 10 Products	460 ms
Some KPI Cards	442–468 ms

Power BI Performance Analyzer was used to review report load times. Sampled visuals were below approximately 500 ms, although performance is environment-dependent. A Validation page was used to check calculations and data consistency; technical pages were hidden from primary navigation.

9. Business Insights and Recommendations

The Executive Overview screenshot shows approximately £989.58M in sales, £425.33M in profit and a 42.98% profit margin. Other conclusions should be verified in the final report.

- Product strategy: Prioritise products with strong revenue and profit contribution; investigate weak margins.
- Regional strategy: Compare geographic performance to find improvement opportunities.
- Customer strategy: Identify high-value customer segments and retention opportunities.
- Pricing strategy: Evaluate discount scenarios under fixed-volume assumptions.

10. Challenges and Solutions

Challenge	Solution
Discount label remained at 0%	Used MAX() for the range slicer
Simulated Sales chart did not update	Corrected the DAX discount reference
Needed different metrics in one chart	Implemented Field Parameters
Needed Sales / Profit switching	Implemented bookmarks and buttons
Needed to assess speed	Used Performance Analyzer
Technical pages in navigation	Hid tooltip, drill-through and validation pages